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Electrical wall plate with movably positionable camera

Abstract

An electrical wall plate may include one or more of the following features. A face plate comprising at least one electrical device opening extending through the face plate. At least one box mounting screw opening configured to receive at least one box mounting screw to fasten the face plate to an electrical device. A power connector extending in a direction away from the face plate and configured to connect with the electrical device. A camera integrated with the face plate and in electrical connection with the power connector and configured to operate by power obtained from the electrical device when the power connector is connected to the electrical device. A camera movement control feature may comprise externally-engageable components coupled to the face plate and configured to movably position the camera or a camera lens by pan or tilt to point in different directions.

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Background/Summary

CLAIM OF PRIORITY (1) This application is a continuation of U.S. patent application Ser. No. 17/880,498 to Jeffrey P. Baldwin titled “Electrical Wall Plate with Movably Positionable Camera” filed Aug. 3, 2022, which is a continuation of U.S. patent application Ser. No. 16/856,793 to Jeffrey P. Baldwin titled “Electrical Wall Plate with Movably Positionable Camera” filed Apr. 23, 2020, which application claims the benefit, including the filing date, of provisional patent application No. 62/837,643 to Jeffrey P. Baldwin titled “Electrical Wall Plate with Moveably Positionable Camera” filed Apr. 23, 2019, the entirety of the disclosures of which are incorporated herein by this reference.

TECHNICAL FIELD

(1) The disclosure relates generally to electrical wall plates and, more particularly, to electrical wall plates having a movably positionable camera.

BACKGROUND

(2) An electrical wall plate is a typical device used to prevent access to electrical wires and electrical receptacles therein. Conventional electrical wall plates commonly mount to electrical receptacles via connection by fasteners, such as screws, or by a mechanical fit involving interlocking components designed to secure the wall plate to the receptacle. Recently, some electrical wall plates have incorporated lights or other electronic components, thereby offering additional functionality beyond mere prevention of access to the electrical wires and receptacle. However, such electrical wall plate designs suffer from one or more of cumbersome and unsafe wiring arrangements, bulky design geometries, and infective communication abilities.

SUMMARY

(3) A need exists for an electrical wall plate that securely mounts to an electrical receptacle, incorporates convenient power connection, includes an efficient design geometry integrating one or more electronic components, such as a movably positionable camera, and affords effective communication ability.

(4) According to an aspect of the disclosure an electrical wall plate may comprise at least one electrical outlet face plate comprising at least one electrical device opening extending through the face plate, wherein the at least one electrical device opening is configured to permit first male electrical plug blades of an electrical power plug to extend through the electrical device opening and engage with a first female receiver of the electrical device, and the at least one face plate

comprises a front face plate and a rear face plate, at least one box mounting screw opening configured to receive at least one box mounting screw to fasten the face plate to an electrical device, a power connector comprising second male electrical plug blades extending in a direction away from the face plate and configured to connect with a second female receiver of the electrical device, a camera integrated within the face plate and in electrical connection with the power connector and configured to operate by power obtained from the electrical device when the power connector is connected to the power transmission conductor of the electrical device, and a camera movement control feature comprising externally-engageable components disposed on a side or on a front surface of the face plate and configured to movably position the camera or a camera lens by pan or tilt to point in different directions by directly applied mechanical force or indirectly through an electrical signal to servo motors.

(5) Particular embodiments may comprise one or more of the following features. The camera movement control feature may be positioned in at least one pin-hole or aperture. The camera movement control feature may comprise at least one slider mechanism. The camera movement control feature may comprise at least one dial mechanism. The camera or camera lens may include a zoom functionality configured to modify the range of the field of view. A printed circuit board (PCB) disposed between the front face plate and the rear face plate, the PCB in electrical communication with the camera and the power connector. A memory storage configured to store photographs or video images captured by the camera. A transmitter configured to transmit signals corresponding to video or photo images captured by the camera to a wireless local area network (WLAN) receiver or a wireless personal area network (WPAN) receiver. An external memory port on the electrical wall plate configured to receive and communicate with a removable external memory device.

(6) According to an aspect of the disclosure, an electrical wall plate may comprise a face plate comprising at least one electrical device opening extending through the face plate, at least one box mounting screw opening configured to receive at least one box mounting screw to fasten the face plate to an electrical device, a power connector extending in a direction away from the face plate and configured to connect with the electrical device, a camera integrated with the face plate and in electrical connection with the power connector and configured to operate by power obtained from the electrical device when the power connector is connected to the electrical device, and a camera movement control feature comprising externally-engageable components coupled to the face plate and configured to movably position the camera or a camera lens by pan or tilt to point in different directions.

(7) Particular embodiments may comprise one or more of the following features. The camera movement control feature may be positioned in at least one pin-hole or aperture disposed on a side or on a front surface of the face plate. The camera movement control feature may comprise at least one slider mechanism disposed on a side or on a front surface of the face plate. The camera movement control feature may comprise at least one dial mechanism disposed on a side or on a front surface of the face plate. The camera movement control feature may comprise a ball and socket joint. The power connector may comprise male electrical plug blades, direct wiring, spring tabs, or a universal serial bus (USB) connector. An external memory port on the electrical wall plate configured to receive and communicate with a removable external memory device.

(8) According to an aspect of the disclosure, a method of capturing video images from an electrical wall plate may comprise providing an electrical wall plate having at least one opening extending through the electrical wall plate, wherein the electrical wall plate further includes a power connector, a printed circuit board in electrical connection with the power connector, and a video camera integrated within the electrical wall plate and in electrical connection with the printed circuit board, movably positioning the camera with a camera movement control feature to point the camera in a desired direction, powering the camera with electricity obtained through an electrical connection between the power connector and an electrical device, and capturing video images by

the camera.

(9) Particular embodiments may comprise one or more of the following features. Storing the captured video images in a memory storage device electrically connected to the printed circuit board of the electrical wall plate. Transmitting the video images via a transmitter electrically connected to the printed circuit board of the electrical wall plate, wherein the transmitter transmits signals to a wireless local area network (WLAN) receiver or a wireless personal area network (WPAN) receiver. A button switch on the electrical wall plate configured to facilitate pairing of the transmitter with a wireless receiver. An electrical receptacle or a light switch. The power connector may comprise male electrical plug blades, direct wiring, spring tabs, or a universal serial bus (USB) connector.

(10) Aspects and applications of the disclosure presented here are described below in the drawings and detailed description. Unless specifically noted, the words and phrases in the specification and the claims be given their plain, ordinary, and accustomed meaning to those of ordinary skill in the applicable arts. The inventors are fully aware that they can be their own lexicographers if desired. The inventors expressly elect, as their own lexicographers, to use only the plain and ordinary meaning of terms in the specification and claims unless they clearly state otherwise and then further, expressly set forth the “special” definition of that term and explain how it differs from the plain and ordinary meaning. Absent such clear statements of intent to apply a “special” definition, it is the inventors' intent and desire that the simple, plain and ordinary meaning to the terms be applied to the interpretation of the specification and claims.

(11) The inventors are also aware of the normal precepts of English grammar. Thus, if a noun, term, or phrase is intended to be further characterized, specified, or narrowed in some way, then such noun, term, or phrase will expressly include additional adjectives, descriptive terms, or other modifiers in accordance with the normal precepts of English grammar. Absent the use of such adjectives, descriptive terms, or modifiers, it is the intent that such nouns, terms, or phrases be given their plain, and ordinary English meaning to those skilled in the applicable arts as set forth above.

(12) Further, the inventors are fully informed of the standards and application of the special provisions of 35 U.S.C. § 112(f). Thus, the use of the words “function,” “means” or “step” in the Detailed Description or Description of the Drawings or claims is not intended to somehow indicate a desire to invoke the special provisions of 35 U.S.C. § 112(f), to define the invention. To the contrary, if the provisions of 35 U.S.C. § 112(f) are sought to be invoked to define the inventions, the claims will specifically and expressly state the exact phrases “means for” or “step for”, and will also recite the word “function” (i.e., will state “means for performing the function of [insert function]”), without also reciting in such phrases any structure, material or act in support of the function. Thus, even when the claims recite a “means for performing the function of . . .” or “step for performing the function of . . .,” if the claims also recite any structure, material or acts in support of that means or step, or that perform the recited function, then it is the clear intention of the inventors not to invoke the provisions of 35 U.S.C. § 112(f). Moreover, even if the provisions of 35 U.S.C. § 112(f) are invoked to define the claimed aspects, it is intended that these aspects not be limited only to the specific structure, material or acts that are described in the preferred embodiments, but in addition, include any and all structures, materials or acts that perform the claimed function as described in alternative embodiments or forms of the disclosure, or that are well known present or later-developed, equivalent structures, material or acts for performing the claimed function

(13) The foregoing and other features, advantages and construction of the present disclosure will become more readily apparent and fully appreciated from the following more DETAILED DESCRIPTION of the particular embodiments and in conjunction with the accompanying DRAWINGS.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) To further clarify the above and other advantages and features of the present disclosure, some of the embodiments will be described in detail, with reference to the following figures, wherein like designations denote like members. It is appreciated that these figures depict only illustrated embodiments pertinent to the present disclosure and are therefore not to be considered limiting of inventive scope. As such, the present disclosure will be set forth and explained with additional specificity and detail through the use of the accompanying drawings in which:

- (2) FIG. 1A illustrates a front left perspective view of an embodiment of an electrical wall plate comprising an integral electronic component or camera;
- (3) FIG. 1B illustrates a front right perspective view of an embodiment of an electrical wall plate comprising an integral electronic component or camera;
- (4) FIG. 2A illustrates a top view of the electrical wall plate of FIGS. 1A and 1B;
- (5) FIG. 2B illustrates a left side view of the electrical wall plate of FIGS. 1A and 1B;
- (6) FIG. 2C illustrates a front view of the electrical wall plate of FIGS. 1A and 1B;
- (7) FIG. 2D illustrates a right side view of the electrical wall plate of FIGS. 1A and 1B;
- (8) FIG. 3 illustrates an exploded perspective view of the electrical wall plate of FIGS. 1A and 1B, as well as an embodiment of an electrical device, an embodiment of an electrical box, and an embodiment of a removable external memory storage device;
- (9) FIG. 4 illustrates a perspective view of another embodiment of an electrical wall plate comprising an integral electronic component or camera;
- (10) FIG. 5 illustrates a perspective view of an embodiment of an electrical plug;
- (11) FIG. 6A illustrates a top view of the electrical wall plate of FIG. 4;
- (12) FIG. 6B illustrates a left side view of the electrical wall plate of FIG. 4;
- (13) FIG. 6C illustrates a front view of the electrical wall plate of FIG. 4;
- (14) FIG. 6D illustrates a right side view of the electrical wall plate of FIG. 4;
- (15) FIG. 7 illustrates an exploded perspective view of the electrical wall plate of FIG. 4, as well as an embodiment of an electrical device, an embodiment of an electrical box, and an embodiment of a removable external memory storage device;
- (16) FIG. 8A illustrates a left perspective view of another embodiment of an electrical wall plate comprising an integral electronic component or camera;
- (17) FIG. 8B illustrates a right perspective view of another embodiment of an electrical wall plate comprising an integral electronic component or camera;
- (18) FIG. 9A illustrates a top view of the electrical wall plate of FIGS. 8A and 8B;
- (19) FIG. 9B illustrates a left side view of the electrical wall plate of FIGS. 8A and 8B;
- (20) FIG. 9C illustrates a front view of the electrical wall plate of FIGS. 8A and 8B;
- (21) FIG. 9D illustrates a right side view of the electrical wall plate of FIGS. 8A and 8B;
- (22) FIG. 10 illustrates an exploded perspective view of the electrical wall plate of FIGS. 8A and 8B, as well as an embodiment of an electrical device, an embodiment of an electrical box, and an embodiment of a removable external memory storage device;
- (23) FIG. 11A illustrates a left perspective view of another embodiment of an electrical wall plate comprising an integral electronic component or camera;
- (24) FIG. 11B illustrates a right perspective view of another embodiment of an electrical wall plate comprising an integral electronic component or camera;
- (25) FIG. 12A illustrates a top view of the electrical wall plate of FIGS. 11A and 11B;
- (26) FIG. 12B illustrates a left side view of the electrical wall plate of FIGS. 11A and 11B;
- (27) FIG. 12C illustrates a front view of the electrical wall plate of FIGS. 11A and 11B;
- (28) FIG. 12D illustrates a right side view of the electrical wall plate of FIGS. 11A and 11B;

(29) FIG. 13 an exploded perspective view of the electrical wall plate of FIGS. 10A and 10B, as well as an embodiment of an electrical device, an embodiment of an electrical box, and an embodiment of a removable external memory storage device;

(30) FIG. 14 illustrates a schematic view of an embodiment of an electrical circuit operably corresponding to an embodiment of an electrical wall plate comprising an integral electronic component;

(31) FIGS. 15A-15E illustrate various views of an embodiment of an electrical wall plate comprising an integral electronic component or camera;

(32) FIG. 16 illustrates a rear perspective view of an embodiment of an electrical wall plate comprising direct wiring for a power connector;

(33) FIG. 17 illustrates a rear perspective view of an embodiment of an electrical wall plate comprising spring tabs for a power connector; and

(34) FIGS. 18A and 18B illustrate a rear and front exploded perspective views of an embodiment of an electrical wall plate comprising universal serial bus (USB) or low voltage connector for a power connector.

DETAILED DESCRIPTION

(35) A detailed description of the hereinafter described embodiments of the disclosed apparatuses, systems and methods are presented herein by way of exemplification and not limitation with reference to the Figures listed above. Although certain embodiments are shown and described in detail, it should be understood that various changes and modifications may be made without departing from the scope of the appended claims. The scope of the present disclosure will in no way be limited to the number of constituting components, the materials thereof, the shapes thereof, the relative arrangement thereof, etc., and are disclosed simply as an example of embodiments of the present disclosure.

(36) Attempts have been made to provide an electrical wall plate including integrated electronic components, but those attempts have been structurally cumbersome, functionally inefficient and cost prohibitive. Referring to the drawings, FIGS. 1A and 1, respectively, depict left and right perspective views of an embodiment of an electrical wall plate **100** comprising an integral electronic component, such as a camera **140**, that solves the aforementioned problems. As depicted, the electrical wall plate **100** may include a front plate face **110**. An electronic device opening **115** may extend through the front plate face **110** of the electrical wall plate **100**. In addition, the electrical wall plate may include a housing **130**. The housing **130** may house electronic components, such as an electrical power transformer, a transmitter, such as an antenna, a printed circuit board, a memory storage component, a battery, a microphone, a speaker, a sensor or multiple sensors, such as a smoke detector or a carbon monoxide detector, a communications port for connecting an external electronic device, a fragrance storage and release element, and/or a mechanical element such as a motor or a structural feature configured to movably extend from the housing **130**, and/or other like components.

(37) Embodiments of an electrical wall plate **100** may include an integral electronic component, such as a camera **140**. Embodiments of the camera **140** may be configured to be movable so that the camera **140** and/or a lens **140a** of the camera **140** may be positioned to point in different directions. Moreover, the camera **140** may include a lens having pan, tilt and zoom functionality, wherein the pan, tilt and zoom functionality may be facilitated by motors, servos, magnets, fluidic pressure devices, heated elements, pneumatic elements, mechanical movement components configured to be engaged and operated by a user, or other features or combinations thereof facilitating movement of the camera **140** and/or the lens of the camera **140**. One or more box mounting screws **152a** and **152b**, may extend through the electrical wall plate **100** and be positioned on the electrical wall plate **100** so as to correspond to similar openings or mechanical features of an electrical device **10**, such as a light switch or and electrical receptacle, or socket fixture. The box mounting screws **152a** and **152b** may facilitate mounting of the electrical wall

plate **100** to the electrical device **10** and also to an electrical box **3**. Depending on the type of electrical device **10** to which the electrical wall plate **100** is mounted, an adapter **190** may be included to facilitate structural correspondence with the electrical device. For example, the adapter **190** may be shaped create an electrical device opening **115** having a shape corresponding with the geometry of a common duplex-type electrical receptacle device **10**. Additionally, embodiments of an electrical wall plate **100** may include a light **170**, such as an LED, as well as a button switch **175**. The button switch **175** may be positioned for manual accessibility and operation. Still further, the light **170** may operate with a sensor that may be positioned in the button switch **175**, wherein the sensor senses ambient light and is configured to send a signal to turn the light on when the sensed ambient light is below a predetermined brightness.

(38) Embodiments of an electrical wall plate **100** may include camera movement control features **141**, such as horizontal movement pin-hole control features **141a** and **141b** and vertical movement pin-hole control features **142a** and **142b**. The camera movement control features may be engaged by or otherwise manipulated by a user to move the camera **140** and/or a lens **140a** of the camera **140**, so that it points in a desired direction. For example, horizontal movement pin-hole control features **141a** and **141b** may be located on the left side or edge of the electrical wall plate **100**. They may be sized so that a pin or a paperclip may be inserted to engage mechanical movement components and/or electrical servo motor switches accessible through the pin-holes. As such, a user who desired to move the camera **140** or lens of the camera **140** horizontally to the left may insert a pin, paperclip, or other slender instrument into the pin-hole control feature **141a** to actuate movement of the camera or lens horizontally to the left. Likewise, if the user wanted to move the camera or lens horizontally to the right the user would insert a pin or paperclip into the pin-hole control feature **141b** to engage the camera movement control features and effectuate movement of the camera **140** and/or lens of the camera **140** horizontally to the right.

(39) In a similar fashion, vertical movement pin-hole control features **142a** and **142b** may be located on the top side or edge of the electrical wall plate **100**. They may also be sized so that a pin, paperclip, or suitable instrument may be inserted to engage mechanical movement components and/or electrical servo motor switches accessible through the pin-holes. As such, a user who desired to move the camera **140** or lens of the camera **140** vertically upward may insert a pin or paperclip into the pin-hole control feature **142a** to actuate movement of the camera or lens in an upward direction. Likewise, if the user wanted to move the camera or lens to point more vertically downward the user would insert a pin or paperclip into the pin-hole control feature **142b** to engage the camera movement control features and effectuate movement of the camera **140** and/or lens of the camera **140** vertically downward.

(40) Movement of the camera **140** or camera lens **140a** can be by directly applied mechanical force, such as through control features **141**, or indirectly through an electrical signal to servo motors. Furthermore, while particular configurations and positions of aperture controls **141a**, **141b**, **142a**, and **142b** have been described with respect to particular camera movements, reverse positioning and reverse movements, as well as other configurations can also be used, as would be appreciated by a person having ordinary skill in the art.

(41) An external memory storage port **188** may also be located on the electrical wall plate **100**. As particularly depicted in FIG. **1B**, the external memory storage port, external memory port, or memory storage **188** may be located on a side or edge, such as the right side or edge. of the electrical wall plate **100**, or at any suitable location along an edge or on a face of the wall plate **100**. The external memory storage port **188** may be configured to receive and facilitate electrical communication with a corresponding external memory storage device.

(42) With further reference to the drawings, FIGS. **2A-2D** depict top, left side, front, and right side views, respectively, of the electrical wall plate of FIGS. **1A** and **1B**. As can be seen, embodiments of an electrical wall plate **100** may comprise a rear face plate **120** opposing the front face plate **110**. Furthermore, embodiments of an electrical wall plate **100** may mechanically and electrically mount

to an electrical device **10**, such as a light switch, electrical receptacle, socket fixture, or common duplex-type electrical receptacle device. The electrical device may be configured and structured so that it mounts to, and substantially within, an electrical box **3**. The electrical box **3** may be located proximally closer to the rear face plate **120** of the electrical wall plate **100** than to the front plate face **110** when the electrical device **10** is mounted therein and the electrical wall plate **100** is mounted to the electrical device **10**.

(43) Various components may be optionally integrated within various embodiments of the electrical wall plate **100**, examples of which are illustrated in the exploded perspective view depicted of FIG. **3**. The electrical wall plate **100** may be electrically connected to, and operate with power or electrical current passing through, a power connector **160** to allow the camera **14** to receive power from the electrical device **10**, without relying on batteries or other power source. The power connector **160** may be conductive and configured to electrically couple or connect with power from the electrical device or electrical receptacle **10**, so as to receive a continuous or continuing source of power. The power connector **160** may comprise male electrical plug blades or second male electrical plug blades **160a** and **160b** extending in a direction away from the face plate **102**, and configured to connect with a female receiver or opening, or second female receiver or opening **17a**, **17b** of the electrical device **10**. The electrical plug blades **160a** and **160b** may perpendicularly protrude beyond, and away from, the rear face plate **120**, as well as extend away from the front plate face **110**. As such, the electrical device **10** may allow for a power plug **1** coupled to an electrical cord **2** (see, e.g., FIG. **5**) to extend through the electronic device opening **115** and couple with the first female receivers **16a**, **16b** of the electrical device **10** with male electrical plug blades **6a**, **6b**, and **6c**. While, or at a same time as, power plug **1** is coupled to the electrical device **10**, the power connector **160** with its male electrical plug blades **160a** and **160b** may be coupled to the second female receivers or openings **17a**, **17b** of the electrical device **110**, allowing for both access to the electrical device **10** to power the camera **140** and simultaneously allowing for other devices or electrical cords to be plugged into the electrical device **10** without interfering with the operation of electrical device **10**, or requiring any additional adapters.

(44) As further illustrated in FIG. **3**, embodiments of an electrical wall plate **100** may include a printed circuit board (PCB) **180**. The PCB **180** may be disposed within the wall plate **100**, such as between the front face plate **110** and the rear face plate **120**. The PCB **180** may include an electrical device opening **115b** corresponding to, and aligned with, the electrical device openings **115a** and **115c** extending through the front face plate **110** and the rear face plate **120**, respectively, of the electrical wall plate **100**. The three openings **115a**, **115b**, and **115c** may together comprise an electrical device opening **115** through the entire electrical wall plate assembly **100**, including the at least one face plate **102**, which comprises front face plate **110** and rear face plate **120**.

(45) The wall plate **100** may also comprise plate mounting screw openings **151** configured to align with box mounting screw openings **13** and to receive box mounting screws **152**. The one or more box screw openings **150**, such as an upper box screw openings **150a** and a lower box screw openings **150b**, may be configured to operatively correspond with box mounting screw openings **13**, such as upper box mounting screw opening **13a** and lower box mounting screw openings **13b**, respectively. The plate mounting screw openings **151** extend through the entirety of electrical wall plate **100** (such as through front face plate **110** and rear face plate **120**) and may also be disposed through the PCB **180**.

(46) The one or more box mounting screws **152a** and **152b** may couple to the electrical wall plate **100** to the electrical device **10**, such as a light switch or an electrical receptacle. The one or more box mounting screw openings **152a** and **152b** may be inserted in, and pass through, the one or more plate mounting screw openings **151a** and **151b**, through one or more corresponding openings **15a** and **15b** of an electrical device, such as an electrical device **10**, and also through openings **13a** and **13b** of electrical box **3**, to facilitate fastening of the electrical wall plate **100** to both the electrical device **10**, such as a light switch or an electrical receptacle, and the electrical box **3**.

(47) Referring still to FIG. 3, the PCB **180** may be in electrical communication with the camera **140** and the power connector **160**. Embodiments of the camera **140** may operate with a camera opening **144** disposed through the face plate **102** or the front face plate **110**. A camera interface component **149** on the PCB **180** may couple the camera **140** to the PCB **180**, as well as provide power and retrieve images and video. The camera **140** may be configured to pan, tilt, zoom, and otherwise move or adjust to a desired orientation for the capture of images and video. The camera movement control features **141**, such as the horizontal and vertical pin-hole control features **142a** and **142b** may be located and configured to engage one or more of the camera **140**, the camera lens **140a**, the camera interface component **149**, the PCB **180**, or portions thereof, when the electrical wall plate **100** is mounted to and electrical device **10**. In addition, embodiments of the camera **140** may physically and electrically connect to the camera interface component **149** of the PCB **180**, wherein the camera interface component **149** may comprise servo motors. Thus, control elements of the PCB **180** may control functionality of the camera **140**, in conjunction with user control via the camera movement control features **141**, such as pin-hole control features **141a**, **141b**, **142a**, and **142b**. The camera **140** may be in electrical connection with the power connector **160** and the PCB **180** and may include powered mechanical elements associated with panning, tilting, and/or zooming. In addition, non-powered mechanical elements may also be provided to facilitate movement of the camera **140** to a desired position, when a user controls position via the camera movement control features **141**, such as pin-hole control features **141a**, **141b**, **142a**, and **142b**. Thus, a camera movement control feature **141** may comprising externally-engageable components disposed on a side or on a front surface of the face plate **102**, such as the front face plate **110**, and be configured to movably position the camera **140** or a camera lens **140a** by pan or tilt to point in different directions, whether by direct mechanical force or indirectly through an electrical signal to servo motors **149**.

(48) Embodiments of the electrical power connector **160** may include a plurality of connecting features, such as electrical plug blades **160a** and **160b**. Those of ordinary skill in the relevant art will recognize the electrical plug blades, such as electrical plug blades **160a** and **160b**, may be structures having standardized dimension and conductivity in compliance with common electrical devices, such as plug and receptacle features having standardized dimensions associated with various geographic regions. For instance, as depicted, the electrical plug blades **160a** and **160b** are structured in accordance with typical plug blade structure in the United States and Canada. However, plug blades, or rather electrical power connector **160** features of embodiments of an electrical wall plate **100** may be structured in accordance with other geographic areas, such as continental Europe, the UK, Australia, China, Japan and any other geographic area having known standardized structure and performance requirements. Likewise, those of ordinary skill in the art will recognize that the component elements of the electrical wall plate **100** may be configured for operation applicable to any voltage, amperage, frequency, wattage, and/or other like electromagnetic characteristic commensurate with standardized functional requirements pertaining to any geographical area. The PCB **180** and other electronic components, such as the camera **140** may function with assistance of known conversion elements, such as operation by transformers, resistors, capacitors, diodes, transistors, A/D converters, integrated circuits and/or other like basic electronic components.

(49) The PCB **180** may facilitate signal processing, device control (electrically, mechanically, or both), signal transmission, data manipulation and storage, user interface and control, and other like functionality. For example, a transmitter **185**, such as a powered antenna, may be in electrical communication with the PCB **180**. Moreover, an integrated memory storage device **187** may also be in electrical communication with the PCB. The integrated memory storage device **187** may be a permanent hard-wired structure. In addition, the PCB may be in electrical communication with a removable external memory storage device **189**, such as a memory card, such as a SD card. The removable external memory storage device **189** may physically and electrically mate with and be

received by an external memory storage port **188**, wherein the external memory storage port **188** may also be located on the electrical wall plate **100** and be in electrical communication with the PCB **180**.

(50) Devices, in communication with the PCB **180**, may operate with electronic components including other separate PCBs, and/or may facilitate functionality of electronic components, such as the camera **140**. For instance, still images or video captured by the camera **140** may be processed through the PCB **180** and directed for storage in the integrated memory storage device **187** and/or in the removable external memory storage device **189**. Furthermore, video images captured by the camera **140** may be processed through the PCB **180** and directed for wireless transmission by the transmitter **185** to an external receiver. Such wireless transmission may include signal transmissions by the transmitter **185** to a wireless local area network receiver (WLAN), such as a Wi-Fi® receiver, signal transmissions by the transmitter **185** to a wireless personal area network receiver (WPAN), such as a Bluetooth® receiver, and/or the wireless transmissions may comprise other like electromagnetic signal transmissions. In addition, it is contemplated that the transmitter **185** may operate via the infrared spectrum, the ultraviolet spectrum, via visible laser light projections, via sonic or ultrasonic transmission, and/or by any other means of operable wireless signal transmission. Still further, the transmitter **185** may transmit over-wire signals, such as analog and/or digital signals passed through the power connector **160** to the electrical power lines connected to the electrical device to which the electrical wall plate **100** may be physically and/or electrically connected. Embodiments of an electrical wall plate **100** may also include a wireless receiver to receive wireless transmissions such as control signals sent from an external source to control operation of the camera **140** via pan and tilt motors.

(51) A button switch **175** may be in electrical communication with the PCB **180**. The button switch **175** may facilitate pairing of the transmitter **185** with a wireless receiver. A plurality of button switches, such as button switch **175**, may be integrated into an embodiment of the electrical wall plate **100**. In addition, a light **170**, such as a light emitting diode (LED), may be in electrical communication with the PCB **180**. The light **170** may function to visually represent operation of one or more electronic components of the electrical wall plate **100**. For instance, the light **170** may blink while the transmitter **185** is pairing with an external receiver, to indicate that the pairing process is properly occurring, it may emit a steady color to indicate normal functionality, it may provide a series of pulses corresponding to user programming instructions and user interfacing, it may turn on during operation of camera **140** to indicate that the camera **140** is properly functioning, and/or the light **170** may be utilized for any other means of information conveyance and user-engagement.

(52) With further reference to the drawings, FIG. **4** depicts a perspective view of another embodiment of an electrical wall plate **200** comprising the integral electronic component, or camera **140**. Many structural and functional aspects of an embodiment of an electrical wall plate **200** may be similar to corresponding structural and functional aspects of an embodiment of an electrical wall plate **100**. Moreover, the perspective view associated with the elemental features depicted in FIG. **4** has been presented in a manner similar to FIG. **1B**, and those in the art will appreciate that structural features and configurations corresponding to the electrical wall plate embodiment **100** of FIG. **1B** likewise correspond to structural features depicted in FIG. **4**. In addition, numeration of component features has been similarly set forth, wherein an embodiment of an electrical wall plate **200** may include a front plate face **110**, a rear plate face **120**, and an electrical device opening **115** that may extend through the front plate face **110** and rear plate face **120** of the electrical wall plate **200**. The electrical wall plate **200** may also include a housing **130** that may house electronic components. Similarly, embodiments of an electrical wall plate **200** may operate with box mounting screws **152a** and **152b**, that extend through the electrical wall plate **200**, as well as an adapter **190** configured to adapt the electrical wall plate **200** to a corresponding structural configuration of an electrical device to which the electrical wall plate **200** will be

mounted. The electrical wall plate **200** may include integral electronic components such as a light **170**, a button switch **175**, and an external memory storage port **188**. Furthermore, an electrical wall plate **200** may include a power connector **160** configured to conductively engage an electrical device, such as by electrical plug blades **160a** and **160b**, or by other conductive elements provided to engage the power transmission features of the electrical device.

(53) Embodiments of an electrical wall plate, such as electrical wall plates **100** and **200**, as well as electrical wall plates **300** and **400** described further herein, and the like, may be configured to connect with be electrically connected to, and operate with power or electrical current passing through, a power connector **160** to allow the camera **14** to receive power from the electrical device **10**, without relying on batteries or other power source. The power connector **160** may be conductive and configured to electrically couple or connect with power from the electrical device or electrical receptacle **10**, so as to receive a continuous or continuing source of power. The power connector **160** may comprise male electrical plug blades or second male electrical plug blades **160a** and **160b** extending in a direction away from the face plate **102**, and configured to connect with a female receiver or opening, or second female receiver or opening **17a**, **17b** of the electrical device **10**. The electrical plug blades **160a** and **160b** may perpendicularly protrude beyond, and away from, the rear face plate **120**, as well as extend away from the front plate face **110**. As such, the electrical device **10** may allow for a power plug **1** coupled to an electrical cord **2** (see, e.g., FIG. 5) to extend through the electronic device opening **115** and couple with the first female receivers **16a**, **16b** of the electrical device **10** with male electrical plug blades **6a**, **6b**, and **6c**. While, or at a same time as, power plug **1** is coupled to the electrical device **10**, the power connector **160** with its male electrical plug blades **160a** and **160b** may be coupled to the second female receivers or openings **17a**, **17b** of the electrical device **110**, allowing for both access to the electrical device **10** to power the camera **140** and simultaneously allowing for other devices or electrical cords to be plugged into the electrical device **10** without interfering with the operation of electrical device **10**, or requiring any additional adapters.

(54) Embodiments of the electrical box **3** may reside on the side of the electrical wall plate **200** associated with the rear plate face **120**, while the camera **140** may reside on the side of the electrical wall plate **200** associated with the front plate face **110**. The electrical device **10**, such as an electrical receptacle, may reside substantially within the electrical box **3**, and may include electrical connection features that are accessible when the electrical box **3** is mounted to the electrical wall plate **100**.

(55) In contrast to an electrical wall plate **100**, embodiments of an electrical wall plate **200** may include a camera **140** that is adjustable via user interface with front plate face aperture controls **243a-d**. The top, left side, front, and right side view depictions of FIGS. **6A-6D** further illustrate the differing structure and functionality of an electrical wall plate **200**. For example, as illustrated, the position adjustment of camera **140** may be facilitated by user engagement with aperture controls located only on the front plate face **110** of the electrical wall plate **200**. Movement control features, such as front plate face aperture controls **243a-d**, may be engaged by or otherwise manipulated by a user to move the camera **140** and/or a lens of the camera **140**, so that it points in a desired direction. For example, front plate face aperture controls **243b** and **243d** may facilitate movement vertically up and down of the camera **140** and/or lens of the camera **140**. The front face plate aperture controls **243a** and **243c** may facilitate movement horizontally left and right of the camera **140** and/or a lens of the camera **140**. Like the pin-hole control features **141a-d** of electrical wall plate **100**, the front plate face aperture controls **243a-d** may be sized so that a pin or a paperclip may be inserted to engage mechanical movement components and/or electrical servo motor switches accessible through the apertures. As such, a user who desired to move the camera **140** or lens of the camera **140** horizontally to the left may insert a pin or paperclip into the front face plate aperture control **243a** to actuate movement of the camera or lens horizontally to the left. Likewise, if the user wanted to move the camera or lens horizontally to the right the user would

insert a pin or paperclip into the front face plate aperture control **243c** to engage the camera movement control features and effectuate movement of the camera **140** and/or lens of the camera **140** horizontally to the right. In a similar fashion, vertical movement front plate face aperture controls **243b** and **243d** may also be sized so that a pin or a paperclip may be inserted to engage mechanical movement components and/or electrical servo motor switches accessible through the apertures. As such, a user who desired to move the camera **140** or lens of the camera **140** vertically upward may insert a pin or paperclip into the front face plate aperture control **243b** to actuate movement of the camera or lens in an upward direction. Likewise, if the user wanted to move the camera **140** and/or a lens of the camera **140** to point more vertically downward the user would insert a pin or paperclip into the front face plate aperture control **243d** to engage the camera movement control features and effectuate movement of the camera **140** and/or lens of the camera **140** vertically downward.

(56) Movement of the camera **140** or camera lens **140a** can be by directly applied mechanical force or indirectly through an electrical signal to servo motors through camera movement control features **141**, such as face plate aperture controls **243a**, **243b**, **243c**, and **243d**. Furthermore, while particular configurations and positions of aperture controls **243** have been described with respect to particular camera movements, reverse positioning and reverse movement or other configurations can also be used, as would be appreciated by a person having ordinary skill in the art.

(57) An external memory storage port **188** may also be located on the electrical wall plate **200**. As particularly depicted in FIG. **6D**, the external memory storage port **188** may be located on the right side or edge of the electrical wall plate **200**. The external memory storage port **188** may be configured to receive and facilitate electrical communication with a corresponding external memory storage device, such as memory storage device **189**.

(58) With further reference to the drawings FIG. **7** depicts an exploded perspective view of the electrical wall plate **200** of FIG. **4**, as well as an embodiment of an electrical device **10**, an embodiment of an electrical box **3**, and an embodiment of a removable external memory storage device **189**. Like embodiments of an electrical wall plate **100**, embodiments of an electrical wall plate **200** include integral electronic components, such as the camera **140**, the light **170** and the button switch **175**, that may be in electrical connection with a power connector, such as power connector **160** having electrical plug blades **160a** and **160b**, and may be configured to operate by electrical power obtained from the electrical device **10**, when the power connector **160** is connected to the power transmission conductor, such as female receivers **16a** and **16b** and/or wire mounts **18a** and **18b**, of the electrical device **10**. The female receivers **16a** and **16b** may be configured to receive and electrically engage the electrical plug blades **160a** and **160b**, when the electrical wall plate **200** is mounted on the electrical device **10**. Those of ordinary skill in the art will recognize that electrical connection between the electrical wall plate **200** and the electrical device **10** may be operatively effectuated by connection with the female receivers **16a** and **16b** and/or the wire mounts **18a** and **18b**, with each connection arrangement offering different advantages. For instance, electrical connection between the power connector **160**, such as electrical plug blades **160a** and **160b**, and the female receivers **16a** and **16b** may happen commensurate with the engagement of the electrical wall plate **200** with the electrical device **10**, as the electrical wall plate **200** is mounted to the electrical device **10**. A direct wired connection, utilizing a wired power connector **160** electrically connected with and physically mounted to the wire mounts **18a** and **18b** may facilitate continuous electrical connection of the electrical wall plate **200** with the electrical device **10**, such as an electrical receptacle, even if the electrical wall plate **200** is spaced apart a distance from the electrical device **10** and not fully physically mounted and/or fastened thereto.

(59) Electronic components integrated into the electrical wall plate **200**, such as the camera **140**, may be powered with electricity obtained through an electrical connection of the electrical wall plate **200** and the electrical device **10** via engagement and reception of the power connector **160**, such as electrical plug blades **160a** and **160b**, with the power transmission conductor, such as

female receivers **16a** and **16b**, of the electrical device **10**. The electronic components may operate with a PCB **180**. For instance, a transmitter **185** and internal memory storage **187** may operate with other electrical components, such as the camera **140**, removable external memory storage **189**, or the light **175** via circuitry of the PCB **180**. Embodiments of the camera **140** may operate with a front face opening **144** and a camera interface component **149** on the PCB **180**.

(60) Referring still further to the drawings, FIGS. **8A** and **8B** depict another embodiment of an electrical wall plate **300** mounted to an embodiment of an electrical device **10** and an embodiment of an electrical device **3**. Many structural and functional aspects of an embodiment of an electrical wall plate **300** may be similar to corresponding structural and functional aspects of an embodiment of an electrical wall plate **100** and/or to an embodiment of an electrical wall plate **200**. Moreover, the perspective view associated with the elemental features depicted in FIGS. **8A** and **8B** has been presented in a manner similar to FIGS. **1A**, **1B**, and **4**, and those in the art will appreciate that structural features and configurations corresponding to the electrical wall plate embodiment **100** and/or to the electrical wall plate embodiment **200** likewise correspond to structural features depicted in FIGS. **8A** and **8B** in correspondence with electrical wall plate embodiment **300**. Accordingly, numeration of component features has been similarly set forth, wherein like features are identified by like numeration. Nevertheless, in contrast to electrical wall plate embodiments **100** and/or **200**, embodiments of an electrical wall plate **300** may include a camera **140** that is adjustable via a user interface or camera movement control features **141**, which may comprise one or more slider mechanism controls, such as slider mechanism controls **345** and **346**, located on the side(s) of electrical wall plate **300**.

(61) The differing structure and functionality of an electrical wall plate **300** is further depicted in the top, left side, front, and right side view illustrations set forth in FIGS. **9A-9D**, respectively. Movement control features, such as slider mechanism controls **345** and **346**, may be engaged by or otherwise manipulated by a user to move the camera **140** and/or a lens **140a** of the camera **140**, so that it points in a desired direction. For instance, a vertical slider mechanism control **345** may be located on the left side of the electrical wall plate **300** and may facilitate movement vertically up and down of the camera **140** and/or lens **140a** of the camera **140**. Moreover, a slider mechanism control **346** may be located on the top of the electrical wall plate **300** and may facilitate movement horizontally left and right of the camera **140** and/or a lens **140a** of the camera **140**. A user who desired to move the camera **140** or lens **140a** of the camera **140** vertically upward may engage the slider mechanism control **345** and slide it upward to effectuate movement of the camera **140** and/or a lens of the camera **140** in an upward direction. The distance the slider mechanism control **345** is slid upward may correspond to the degree of upward angular direction that the camera **140** and/or a lens of the camera **140** may be variably positioned. Similarly, if the user wanted to move the camera **140** and/or a lens of the camera **140** so that it points vertically downward, the slider mechanism control **345** may be slid vertically downward until the camera **140** or lens of the camera **140** is correspondingly pointed in the desired downwardly-angled position.

(62) In a like manner, a user who desired to move the camera **140** or lens of the camera **140** horizontally to the left may engage the horizontal slider mechanism control **346** and slide it to the left to effectuate movement of the camera **140** and/or a lens of the camera **140** in left horizontal direction. The distance the slider mechanism control **346** is slid to the left may correspond to the degree of leftward angular direction that the camera **140** and/or a lens **140a** of the camera **140** may be variably positioned. In the same way, if a user wanted to move the camera **140** and/or a lens **140a** of the camera **140** so that it points horizontally toward the right, the slider mechanism control **346** may be slid horizontally to the right until the camera **140** or lens of the camera **140** is correspondingly pointed in the desired rightwardly-angled position. One of ordinary skill in the art will appreciate that appropriate actuation of the slider mechanism may effectuate movement of the camera **140** or a lens **140a** of the camera **140** toward and into any angularly directed position desired by a user with regard to the field of view presented to front face plate **110**.

(63) Movement of the camera **140** or camera lens **140a** can be by directly applied mechanical force or indirectly through an electrical signal to servo motors through camera movement control features **141**, such as dial mechanism controls **447** and **448**. Furthermore, while particular configurations and positions of the dial mechanism controls **447** and **448** have been described with respect to particular camera movements, reverse positioning and reverse movement or other configurations can also be used, as would be appreciated by a person having ordinary skill in the art.

(64) An external memory storage port **188** may also be located on the electrical wall plate **300** and may be configured to receive and facilitate electrical communication with a corresponding external memory storage device, such as memory storage device **189**.

(65) Comparable to embodiments of an electrical wall plate **100** and/or an electrical wall plate **200**, embodiments of an electrical wall plate **300** include integral electronic components, as depicted in FIG. **10**, such as the camera **140**, the light **170** and the button switch **175**, that may be in electrical connection with a power connector, such as power connector **160** having electrical plug blades **160a** and **160b**, and may be configured to operate by electrical power obtained from the electrical device **10**, when the power connector **160** is connected to the power transmission conductor, such as female receivers **17a** and **17b** and/or wire mounts **18a** and **18b**, of the electrical device **10**. The female receivers **17a** and **17b** may be configured to receive and electrically engage the electrical plug blades **160a** and **160b**, when the electrical wall plate **300** is mounted on the electrical device **10**. Again, those of ordinary skill in the art will recognize that electrical connection between the electrical wall plate **300** and the electrical device **10** may be operatively effectuated by connection with the female receivers **17a** and **17b** and/or the wire mounts **18a** and **18b**, with each connection arrangement offering different advantages. Electronic components integrated into the electrical wall plate **300**, such as the camera **140**, may operate with a PCB **180**. For instance, a transmitter **185** and internal memory storage **187** may operate with other electrical components, such as the camera **140**, removable external memory storage **189**, or the light **175** via circuitry of the PCB **180**. Embodiments of the camera **140** may operate with a front face opening **144** and a camera interface component **149** on the PCB **180**.

(66) With further reference to the drawings, another embodiment of an electrical wall plate **400** mounted to an embodiment of an electrical device **10** and an embodiment of an electrical device **3** is depicted FIGS. **11A** and **11B**. Again, many structural and functional aspects of an embodiment of an electrical wall plate **400** may be similar to corresponding structural and functional aspects of embodiments of one or more of the electrical wall plates **100**, **200**, and **300**. Additionally, the perspective view associated with the elemental features depicted in FIGS. **11A** and **11B** has likewise been presented in a manner similar to FIGS. **1A**, **1B**, **4**, **8A**, and **8B**, and those having ordinary skill in the art will appreciate that structural features and configurations corresponding to one or more of the electrical wall plate embodiments **100**, **200** and **300** that similarly correspond to structural features depicted in FIGS. **11A** and **11B** for electrical wall plate embodiment **400**. As such, numeration of component features has been correspondingly set forth, wherein like features are identified by like numeration. Yet, in contrast to electrical wall plate embodiments **100**, **200** and/or **300**, embodiments of an electrical wall plate **400** may include a camera **140** that is adjustable via user interface with one or more camera movement control features **141**, such as dial mechanisms controls like dial mechanism controls **447** and **448**, located on the side(s), face(s) or suitable surfaces of the electrical wall plate **400**.

(67) As shown in the top, left side, front, and right side view illustrations set forth respectively in FIGS. **12A-12D**, camera movement control features **141**, such as dial mechanism controls **447** and **448**, may be actuated and spun by a user to move the camera **140** and/or a lens **140a** of the camera **140**, so that it points in a desired direction. A user may engage the dial mechanism control **447** and spin it upward to effectuate movement of the camera **140** and/or a lens of the camera **140** in an upward direction. The rotational distance the dial mechanism control **447** is spun upward may

correspond to the degree of upward angular direction that the camera **140** and/or a lens **140a** of the camera **140** may be variably positioned. In addition, a user may spin the dial mechanism control **447** rotationally downward until the camera **140** or lens **140a** of the camera **140** is correspondingly pointed in the desired downwardly-angled position. Also, a user may engage the dial mechanism control **448** and spin it rotationally to the left to effectuate movement of the camera **140** and/or a lens **140a** of the camera **140** in left horizontal direction and may spin the dial mechanism control **448** rotationally to the right to effectuate movement of the camera **140** or lens of the camera **140** in a right horizontal direction. The distance the dial mechanism control **448** is spun, either to the right or to the left, may correspond to the variable degree of angular direction that the camera **140** and/or a lens **140a** of the camera **140** may be correspondingly positioned toward and into any angularly directed position desired by a user. In some instances, the camera movement control features **141**, including one or more dial mechanism controls, sliders, or opening/pin holes may be used to increase or decrease zoom of the camera **140** or positioning of the lens **141**, as well as increase or decrease the field of view of the camera **140**.

(68) Movement of the camera **140** or camera lens **140a** can be by directly applied mechanical force or indirectly through an electrical signal to servo motors through camera movement control features **141**, such as face plate aperture controls **243a**, **243b**, **243c**, and **243d**. Furthermore, while particular configurations and positions of aperture controls **243** have been described with respect to particular camera movements, reverse positioning and reverse movement or other configurations can also be used, as would be appreciated by a person having ordinary skill in the art.

(69) An external memory storage port **188** may also be located on the electrical wall plate **400** and may be configured to receive and facilitate electrical communication with a corresponding external memory storage device, such as memory storage device **189**.

(70) Like the depictions of embodiments of an electrical wall plate **100**, **200**, and **300** shown respectively in FIGS. **3**, **7**, and **10**, embodiments of an electrical wall plate **400**, as depicted in FIG. **13**, may include integral electronic components, such as the camera **140**, the light **170** and the button switch **175**, that may be in electrical connection with a power connector, such as power connector **160** having electrical plug blades **160a** and **160b**, and may be configured to operate by electrical power obtained from the electrical device **10**, when the power connector **160** is connected to the power transmission conductor, such as female receivers **17a** and **17b** and/or wire mounts **18a** and **18b**, of the electrical device **10**. Electronic components integrated into the electrical wall plate **400**, such as the camera **140**, may electrically connect with circuitry of a PCB **180** and operate with components such as a transmitter **185** and internal memory storage **187**, external memory storage **189**, and/or a light **175**. Again, embodiments of the camera **140** may operate with a front face camera opening **144** and a camera interface component **149** on the PCB **180**.

(71) Embodiments of an electrical wall plate **100**, **200**, **300**, **400** and the like may be mounted to an electrical box **3**. When mounted to the electrical box **3**, an electronic component, such as a camera **140** and the like may remain highly visible and capable of capturing video images of the environment located in front of the front plate faces **110** of the electrical wall plate(s) **100**, **200**, **300**, **400** and the like. Additionally, the rear plate faces **120** of the electrical wall plate **100**, **200**, **300**, **400** and the like may be flush-mounted to the electrical box **3** and thereby protect the electrical device **10**, such as an electrical receptacle or a light switch, which is housed by the electrical box **3**, from undesired accessibility.

(72) Referring still further to the drawings, FIG. **14** depicts a schematic view of an embodiment of an electrical circuit **2000** operably corresponding to an embodiment of an electrical wall plate, such as electrical wall plates **100**, **200**, **300** and/or **400** comprising an integral electronic component such as cameras **140** and **2040**. A printed circuit board **2080** is electrically connected to the camera, such as camera **2040**, AC electrical power **2010**, an LED light **2070**, a button switch **2075**, and a transmitter antenna **2085**. As shown schematically, when the camera, such as camera **2040**, is electrically powered by electricity, such as AC power **2010**, obtained through electrical connection

of the electrical wall plate, such as electrical wall plate **100**, **200**, **300**, **400**, and the electrical device **10**, the electrically connected PCB **2080** may help facilitate controlled operation of the camera **2040**.

(73) FIGS. **15A-18B** illustrate various views of an embodiment of an electrical wall plate **500** comprising an integral electronic component or camera **140**, similar to electrical wall plates **100**, **200**, **300**, and **400** as described herein with respect to the other figures. A person of ordinary skill in the art (POSA) will understand that like reference numbers refer to like features, and any exemplary features described with respect to electrical wall plates **100**, **200**, **300**, **400**, and **500** may be applied to, or used with, any variation, combination, or permutation of any other electrical wall plate **100**, **200**, **300**, **400**, and **500** shown and described herein, including with respect to various camera movement control features **141** and power connectors **160**.

(74) FIGS. **15A-18B** differ from the previous figures by including a camera **140** formed comprising a ball and socket arrangement **143**. The camera lens **140a** may be disposed within the camera ball **143a**, and the ball **143a** may be coupled to, or partially disposed within, the socket **143b** of the camera **140**. By being configured as a ball and socket structure **143**, camera movement control feature **141** of the ball and socket camera **143** may be the ball **143a**, which may be moveable or positionable within the socket **143b**. As such, the ball and socket structure **143** may allow (or being configured or adapted to allow) the camera lens **140a** to be adjusted, moved, or controlled, by pan or tilt to point in different directions, such as by directly applied mechanical force or indirectly through an electrical signal to servo motors.

(75) FIG. **15A** shows a front perspective view of the electrical wall plate **500**, and also provides an illustration of the electrical device **10** being a light switch **10**, which may be one or more of a toggle light switch, dimmer, turn dial, plunger, flip switch, touch pad, remote wirelessly controlled interface, or any other suitable switch. In some instances, the electrical device **10** may comprise at least one light switch, at least one electrical receptacle, or both, as well as any other suitable arrangement of known electrical devices. FIG. **15B** shows a rear perspective view of an electrical wall plate **500** with the camera **140** hidden from view on the front face plate **110**. FIG. **15C** shows an exploded perspective view of the electrical wall plate **500**, comprising at least one face plate **102**, the PCB **180**, the electrical device **10**, and the electrical box **3**.

(76) FIG. **15D** shows a front profile view of the electrical wall plate **500** with the ball and socket camera **143** disposed at the upper left corner of the wall plate **500**, although any position on the wall plate **500** may be used. FIG. **15D** also shows a section line **15E-15E**, along which the view cross-sectional side or profile view of FIG. **15E** is taken.

(77) FIGS. **16-18B** show various views of different power connectors **160** that may be used with the wall plate **500**, instead of the male electrical plug blades **160a**, **160b** that could also be used the wall plate **500**. As indicated above, the POSA will understand that the various power connectors **160**, including direct wiring **160c**, spring tabs **160d**, and USB connector or low voltage connector **160e** (shown with respect to wall plate **500** in FIGS. **16-18B**) may also be used with any of the other wall plates **100**, **200**, **300**, **400**, or variations thereof.

(78) FIG. **16** illustrates a rear perspective view of an embodiment of an electrical wall plate **500** comprising direct wiring **160c** that is electrically coupled with the camera or electronic component **140**, such as through the PCB or circuit board **180**, and may be coupled to the electrical device **10**. The direct wiring **160c** is coupled with, to, or directly contacts, electrical power transmission conductors **18a** and **18b**, which may be formed as screws or other suitable structures.

(79) FIG. **17** illustrates a rear perspective view of an embodiment of an electrical wall plate **500** comprising spring tabs **160d** for a power connector **160**. The spring tabs **160d** may be electrically coupled with the camera or electronic component **140**, such as through the PCB or circuit board **180**, and may be coupled to the electrical device **10**. The spring tabs **160d** are coupled with, to, or directly contact, electrical power transmission conductors **18a** and **18b**, which may be formed as screws or other suitable structures.

(80) FIGS. 18A and 18B illustrate rear and front exploded perspective views, respectively, of an embodiment of an electrical wall plate 500 comprising USB or low voltage connectors 160e for a power connector 160. FIG. 18A shows an instance in which a portion or first portion of the USB connector 160e, such as a male USB connector, may extend from the PCB 180 in a direction away from the wall plate 500 (similar to male electrical lug blades 160a, 160b). The first portion or male USB connector 160e may be received in, or couple with, a second portion of the USB connection or a USB electrical power transmission receiver or opening 18c formed as part of the electrical device 10. The front perspective view of FIG. 18B provides additional detail of the second portion, female receptacle, or USB electrical power transmission receiver or opening 18c.

(81) While this disclosure has been described in conjunction with the specific embodiments outlined above, it is evident that many alternatives, modifications and variations will be apparent to those skilled in the art. Accordingly, the described and depicted embodiments of the present disclosure as set forth above are intended to be illustrative, not limiting. Various changes may be made without departing from the spirit and scope of the present disclosure, as required by the following claims. The claims provide the scope of the coverage of the present disclosure and should not be limited to the specific examples provided herein.

Claims

1. An electrical wall plate, comprising: at least one electrical face plate comprising at least one electrical device opening extending through the face plate, wherein the at least one electrical device opening is configured to expose a front of an electrical device through the electrical device opening, wherein the at least one face plate comprises a front plate having a socket, a rear plate, and a printed circuit board between the front plate and the rear plate, wherein the at least one electrical device opening extends through the front plate and the rear plate of the at least one electrical face plate; at least one screw opening extending through the face plate configured to receive at least one screw to fasten the face plate to the electrical device; a camera formed as a ball and rotatably coupled with the socket of the front plate, wherein the camera extends forward from the front plate and wherein the camera is operatively coupled to the printed circuit board and configured to rotate within the socket in any direction with respect to the front plate; and a first power connector affixed to and extending away from the printed circuit board through the rear plate, the first power connector configured to mateably couple with a second power connector positioned on the electrical device when the face plate is fastened to the electrical device.
2. The electrical wall plate of claim 1, wherein the camera is configured to rotate with respect to the socket by a directly applied mechanical force.
3. The electrical wall plate of claim 1, wherein the camera is configured to rotate with respect to the socket through an electrical signal to servo motors.
4. The electrical wall plate of claim 1, further comprising a memory storage configured to store photographs or video images captured by the camera and a transmitter configured to communicate the photographs or video images to a remote device.
5. The electrical wall plate of claim 1, further comprising a first pinhole and a second pinhole each extending through the face plate, and each configured to expose an electrical component of the printed circuit board.
6. An electrical wall plate, comprising: at least one electrical face plate comprising at least one electrical device opening extending through the face plate, wherein the at least one electrical device opening is configured to expose a front of an electrical device through the electrical device opening and wherein the at least one face plate comprises a front plate, a rear plate, and an electrical circuit between the front plate and the rear plate; at least one screw opening extending through the face plate configured to receive at least one screw to fasten the face plate to the electrical device; a camera integrated with the face plate and exposed through a camera aperture in the face plate,

wherein the camera is operatively coupled to the electrical circuit; a first pinhole and a second pinhole each extending through the face plate, and each configured to expose an electrical component of the electrical circuit; and a first power connector affixed to and extending away from the electrical circuit through the rear plate, the first power connector configured to couple with a second power connector positioned on the electrical device when the face plate is fastened to the electrical device.

7. The electrical wall plate of claim 6, wherein the front plate has a socket and the camera is formed as a ball and configured to rotate within the socket.

8. The electrical wall plate of claim 7, wherein the camera is configured to rotate with respect to the socket by a directly applied mechanical force.

9. The electrical wall plate of claim 7, wherein the camera is configured to rotate with respect to the socket through an electrical signal to servo motors.

10. The electrical wall plate of claim 6, further comprising a memory storage configured to store photographs or video images captured by the camera and a transmitter configured to communicate the photographs or video images to a remote device.

11. An electrical wall plate, comprising: at least one electrical face plate comprising at least one electrical device opening extending through the face plate, wherein the at least one electrical device opening is configured to expose a front of an electrical device through the electrical device opening and wherein the at least one face plate comprises a front plate, a rear plate, and an electrical circuit between the front plate and the rear plate; at least one screw opening extending through the face plate configured to receive at least one screw to fasten the face plate to the electrical device; a camera rotatably coupled with the socket of the front plate, wherein the camera extends forward from the front plate and is operatively coupled to the electrical circuit; a first pinhole and a second pinhole each extending through the face plate, and each configured to expose an electrical component of the electrical circuit; and a first power connector affixed to and extending away from the electrical circuit through the rear plate, the first power connector configured to couple with a second power connector positioned on the electrical device when the face plate is fastened to the electrical device.

12. The electrical wall plate of claim 11, wherein the camera is integrated with the face plate and exposed through a camera aperture in the face plate.

13. The electrical wall plate of claim 11, wherein the second power connector is positioned on a side of the electrical device distinct from the front of the electrical device.

14. The electrical wall plate of claim 11, wherein the front plate has a socket and the camera is formed as a ball and configured to rotate within the socket.

15. The electrical wall plate of claim 14, wherein the camera is configured to rotate with respect to the socket by a directly applied mechanical force.

16. The electrical wall plate of claim 14, wherein the camera is configured to rotate with respect to the socket through an electrical signal to servo motors.

17. The electrical wall plate of claim 11, wherein the first power connector is a male USB power connector and the second power connector is a female USB power connector.

18. The electrical wall plate of claim 11, further comprising a memory storage configured to store photographs or video images captured by the camera and a transmitter configured to communicate the photographs or video images to a remote device.
